The Fine-Structure Constant as an OPH Pixel Fixed Point

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Abstract

Observer Patch Holography (OPH) is the finite observer-overlap framework introduced in *Observers Are All You Need* [1]. In that framework, one OPH screen cell has two readings. From the outside, the cell has a dimensionless area P. From the inside, it appears as the electromagnetic observation scale available to observers in the encoded world. The two readings meet at

$$P = \varphi + \frac{\sqrt{\pi}}{A_{\mathrm{Th}}(P)}.$$

Solving this self-referential pixel equation with the OPH source map gives a fine-structure fixed point close to the measured low-energy value. Informally, the source-side no-hadron branch starts from the OPH screen-cell rules without fitted parameters; the public Thomson endpoint still requires the separate low-energy hadronic transport closure below. The source-plus-unified-gauge-width value is

$$A^{\mathrm{fp}}_{\alpha_U} = 137.035959500817279952949994585787\ldots.$$

The most precise direct measurement, the 2020 rubidium recoil result, is

$$\alpha^{-1}_{\mathrm{Rb}} = 137.035999206(11),$$

so the OPH output is low by

$$0.000039705182720047050005414213\ldots,$$

about 2.89743×10^{-7} relative in inverse-alpha units. That remaining gap is the part controlled by low-energy quark and gluon physics. At these energies quarks are confined inside hadrons, so the correction cannot be obtained by simply running a perturbative QCD calculation. It requires a non-perturbative hadronic spectral calculation in the same electromagnetic convention as the OPH source map, which is the open step.

What This Paper Contributes

The fine-structure constant is usually treated as an empirical constant. This paper turns it into a branch readout of one OPH screen cell. The known physics is the low-energy Maxwell coupling, charged-lepton running, and low-energy hadronic spectral transport. OPH adds the pixel self-reference: the outside area coordinate P must agree with the inside electromagnetic observation scale read by observers on the same branch.

The source map gets close from OPH source data alone. The remaining gap has an explicit address: the hadronic transport needed in the same electromagnetic convention as the source map. This split is the contribution: a fixed-point equation for the pixel branch, a visible address for the missing low-energy QCD step, and a clear boundary between source theorem, empirical endpoint closure, and measured comparison.

1 Introduction

The goal is to make every mathematical step visible and keep the physics boundary clear.

There is a long history of attempts to understand α from deeper structure. Jentschura and Nandori give a useful survey of first-principles attempts, including beta-function, symmetry, cutoff, and string-inspired ideas [6]. Golden-ratio constructions appear in historical geometric proposals [7] and in recent semi-empirical work on electroweak and flavor mixing [9]. Octonionic and exceptional-algebra approaches also produce values near 1/137 [8]. Recent arXiv papers use symbolic regression and quantum-information criteria to look for structure in Standard Model constants and electroweak parameters [10, 11]. The OPH calculation belongs to this broad search for structure, with one added requirement: the number must arise from a fixed-point map whose input and output are the same local screen cell.

> **Pure source calculation.** The executable source map solves the OPH fixed point from screen-cell data alone. Its no-hadron output is the undressed source/root inverse coupling $\alpha_{\rm root}^{-1} = 136.9948351646\ldots$. Adding the emitted finite-screen unified gauge-width contribution $\alpha_U(P_\star)$ gives the source-side no-hadron near-endpoint $A_{\alpha_U}^{\rm fp} = 137.0359595008\ldots$, close to the laboratory value.
>
> **CODATA/NIST measured endpoint.** For comparison, CODATA/NIST reports $\alpha^{-1}(0) = 137.035999177(21)$. The gap from $A_{\alpha_U}^{\rm fp}$ identifies the remaining low-energy hadronic correction in the same electromagnetic convention as the source map.

Intuitive picture: three readings of one pixel. The three fine-structure numbers are not three unrelated corrections. They are three increasingly observer-facing readings of the same local pixel branch:

$$\text{source pixel} \longrightarrow \text{gauge-width dressed pixel} \longrightarrow \text{measured low-energy value},$$

or numerically,

$$136.9948351646\ldots \longrightarrow 137.0359595008\ldots \longrightarrow 137.035999177(21).$$

The root value $\alpha_{\rm root}^{-1}$ is the naked source/root inverse coupling: the electromagnetic width emitted when one screen cell closes as a geometric self-consistency object before the low-energy charged vacuum is attached. The outside reading is the pixel detuning $(P-\varphi)/\sqrt{\pi}$; the inside reading is the electromagnetic observation strength on the same branch. The fixed point says those two readings must name one cell.

The contribution $\alpha_U(P_\star)$ is the finite-screen unified gauge-width contribution. It is the main high-scale gauge-sector width carried by the same pixel: the weak and color representation content must fit the pixel budget $P/4$, and the D10 heat-kernel closure supplies the executable certificate.

Adding this width to the root gives the source-side no-hadron near-endpoint $A^{\mathrm{fp}}_{\alpha_U}$. It is close to the laboratory endpoint because it supplies almost all of the inverse-alpha gap from the naked source value.

The factor $C_{24,Q}$ is the small endpoint dressing required after the gauge-width contribution is added. It is near one because $\alpha_U(P_\star)$ supplies the large part of the correction. Its excess $\alpha_U(P_\star)(C_{24,Q}-1)$ represents the remaining same-scheme low-energy hadronic endpoint transport. A zero-momentum laboratory photon does not probe an empty charged vacuum; it sees the Ward-projected $\mathrm{U}(1)_Q$ current after charged-lepton vacuum polarization, confined-quark/hadron spectral transport, and finite endpoint matching. In short, the observer measures the fully dressed low-energy coupling, while $A^{\mathrm{fp}}_{\alpha_U}$ is the no-hadron value before that dressing.

The OPH background used here is spread across three papers. The observer-overlap starting point is the synthesis paper [1]. The source map and fixed-point branch are recorded in the compact overlap-consistency paper [2]. The charged-spectrum and particle-structure continuation is recorded in the particle paper [3]. The references give direct GitHub links to those source files.

Why this split matters: the fixed-point equation and the hadronic transport calculation answer different questions. The fixed-point equation says how a local screen cell must close on itself. The hadronic calculation says how the electromagnetic current is transported through low-energy QCD. These are different layers of the calculation. A non-arbitrary endpoint calculation must either evaluate the same-scheme hadronic spectral functional directly, or derive the scheme bridge from OPH source data.

Proposition 1 (Maxwell normalization of the endpoint lane)**.** *On the ordinary electromagnetic branch used by the fixed-point equation, the Ward-projected* $\mathrm{U}(1)_Q$ *channel carries*

$$F_Q = dA_Q, \qquad dF_Q = 0, \qquad d{*}F_Q = g_Q^2(q^2;P){*}J_Q,$$

with

$$g_Q^2(q^2;P) = 4\pi\alpha_{\mathrm{em}}(q^2;P), \qquad A_{\mathrm{Th}}(P) = \alpha_{\mathrm{em}}^{-1}(0;P).$$

Thus the inside reading in the pixel equation is the Thomson-limit Maxwell coupling of the same electromagnetic current.

Proof. The compact OPH reconstruction identifies the unbroken electromagnetic branch as $\mathrm{U}(1)_Q$. On that abelian factor, the compact-gauge curvature reduces to $F_Q = dA_Q$, and the quadratic field action gives $d{*}F_Q = g_Q^2{*}J_Q$. The particle-paper transport theorem then identifies the same $g_Q(q^2;P)$ with the Ward-projected running electromagnetic coupling. Taking $q^2 \to 0$ gives the endpoint quantity $A_{\mathrm{Th}}(P)$ used in the pixel fixed-point equation. □

The hadronic step uses the standard dispersion logic behind hadronic vacuum-polarization work. The relevant comparison data are electromagnetic spectral functions measured through $e^+e^- \to$ hadrons, as used in data-driven running-α and $g-2$ analyses [12, 13, 16]. In this paper, the decimal correction shown above is reported as the difference between the CODATA/NIST comparison value and the OPH pure source calculation. A direct OPH production calculation of the hadronic spectral function remains open.

2 Symbols

Symbol	Definition	Informal meaning
$\mathcal{A}(P)$	Patch algebra assigned to a screen patch P	The local observables available to one finite observer patch.
ω_P	State on $\mathcal{A}(P)$	The local physical description carried by that patch.
$\mathcal{A}(P \cap Q)$	Shared overlap algebra	The observables two neighboring patches can compare.
P	$a_{\text{cell}}/\ell_\star^2$	The dimensionless area of one screen cell in scale-readout units.
a_{cell}	Physical screen-cell area	The area of the local pixel on the holographic screen.
$\ell_\star$	OPH scale-certificate length	The length emitted by the selected no-G scale certificate.
ℓ_P	Planck length	The SI display of $\ell_\star$ after $G_{\text{SI}} = c^3\ell_\star^2/\hbar$.
φ	$(1+\sqrt{5})/2$	The self-similar entropy-balance point.
$\sqrt{\pi}$	Boundary Gaussian normalization width	The conversion factor from pixel displacement to observation width.
$\alpha_{\text{ext}}(P)$	$(P-\varphi)/\sqrt{\pi}$	The outside, geometric reading of the electromagnetic coupling.
$A_{\text{Th}}(P)$	$\alpha_{\text{em}}^{-1}(0; P)$	The inverse electromagnetic coupling at the Thomson limit emitted by the trial pixel.
$\alpha_{\text{in}}(P)$	$1/A_{\text{Th}}(P)$	The inside, electromagnetic reading of the same cell.
E_P	Planck energy	The energy unit associated with the selected scale certificate. The numerical equations use Planck units unless GeV labels are attached.
$M_U(P)$	$E_P \mathrm{e}^{-2\pi} P^{1/6}$	The OPH unification-scale readout emitted by a trial pixel.
$E_{\text{cell}}(P)$	$E_P/\sqrt{P}$	The local cell energy readout.
$\alpha_U(P)$	Unified coupling solved from the heat-kernel closure equation	The coupling at $M_U(P)$ that makes the gauge representation entropy match the pixel.
$\alpha_i(\mu; P)$	Gauge coupling $i = 1, 2, 3$ at scale μ	The hypercharge, weak, and color couplings emitted by the same trial P.
b_i	$(33/5, 1, -3)$	The one-loop beta coefficients for the high-scale running convention used here.

Symbol	Definition	Informal meaning
$m_Z(P)$	Self-consistent electroweak anchor scale	The Z-scale generated by $v(P)$, $\alpha_1(P)$, and $\alpha_2(P)$.
$v(P)$	$E_{\mathrm{cell}}(P)\exp[-2\pi/(\beta_{\mathrm{EW}}\alpha_U(P))]$	The electroweak transmutation scale.
β_{EW}	$N_c+1=4$ for $N_c=3$	The coefficient that controls the exponential drop to the electroweak scale.
$A_Z(P)$	$\alpha_{\mathrm{em}}^{-1}(m_Z^2;P)$	The electroweak-scale electromagnetic anchor.
$\Delta_{\mathrm{lep}}(P)$	Lepton transport contribution to inverse alpha	The exact one-loop e,μ,τ contribution.
$\Delta_{\mathrm{had}}(P)$	Hadronic transport contribution	The low-energy QCD spectral contribution needed for the pure source calculation.
$\Delta_{\mathrm{EW}}(P)$	Electroweak/scheme endpoint remainder	Same-scheme finite remainder needed to connect the source anchor to the endpoint.
$\Delta_{\mathrm{calc}}(P)$	Calculated charged-fermion transport contribution	The exact lepton term plus the screened quark continuation used by the source calculation.
$\Delta^{\mathrm{fp}}_{\mathrm{H,cal}}$	Root-to-CODATA bookkeeping difference	The inverse-alpha amount between the naked root value and the CODATA/NIST central endpoint; it includes the emitted $\alpha_U(P_\star)$ contribution and is not the post-α_U hadronic gap.
$\Delta^{\mathrm{fp}}_{\mathrm{H,req}}$	Required hadronic endpoint correction after $A^{\mathrm{fp}}_{\alpha_U}$	The inverse-alpha amount by which the current OPH output misses the CODATA/NIST central endpoint.
$R_Q(P)$	$A_{\mathrm{Th}}(P)-[A_Z(P)+\Delta_{\mathrm{calc}}(P)]$	The remaining same-scheme hadronic endpoint contribution.
$A^{\mathrm{fp}}_{\alpha_U}$	$\alpha_{\mathrm{root}}^{-1}+\alpha_U(P_\star)$	Source-side no-hadron prediction before the Ward-projected hadronic transport is supplied.
$J_{24,Q}(P),\omega_Q(P),\Xi_Q(P)$	Source-side hadronic spectral objects required for a final endpoint theorem	The Jacobi/Stieltjes spectral payload, normalization, and finite same-scheme remainder that must be emitted without looking at the Thomson target.
P_{C}	CODATA/NIST comparison pixel	The pixel obtained from the CODATA/NIST Thomson endpoint under the OPH outer equation.

3 Shared Collar Convention for χ_ν

The χ_ν protected-reserve theorem uses the public endpoint branch convention

$$P_\chi = P_{\mathrm{C}} = 1.630968209403959\ldots.$$

Its scalar reserve density is

$$\epsilon_{\mathrm{res}} = \frac{P_\chi}{24}$$

only after the same-collar shared-edge budget and one-class $\mathbb{Z}_6$ trace receipts pass in the screen microphysics and susceptibility papers. This does not make $P/4$ a primitive Hilbert-space dimension. In this paper, $P/4$ is the local screen-cell entropy budget used by the electromagnetic fixed-point lane; $P_\chi/24$ is the shared scalar-reserve density used by the χ_ν collar branch.

4 Step 1: The OPH Starting Rule

OPH starts from a finite screen covered by observer patches. If P_1 and P_2 overlap, then their induced states must agree on the shared algebra:

$$\omega_{P_1}|_{\mathcal{A}(P_1\cap P_2)} = \omega_{P_2}|_{\mathcal{A}(P_1\cap P_2)}. \tag{1}$$

Informally: no observer sees the whole world. Physics is what survives when neighboring local descriptions can be checked against each other and made consistent on the parts they share.

The quantitative fine-structure branch applies this rule to a single local screen cell. The cell has one outside description and one inside description. The outside description is geometric. The inside description is electromagnetic. Closure means both descriptions identify the same cell.

Why this step is needed: OPH does not start by assigning constants to nature. It starts by asking what finite observers can compare. A dimensionless constant can be derived only if it is the fixed value of such a comparison.

5 Step 2: The Pixel Variable

Define the local pixel ratio

$$P := \frac{a_{\mathrm{cell}}}{\ell_\star^2}. \tag{2}$$

Here a_{cell} is the screen-cell area and $\ell_\star^2$ is the area scale emitted by the separate OPH gravity readback fixed point.

Informally: P says how large the screen cell is when measured in scale-readout units. It is dimensionless, so it can be compared directly with pure numbers such as φ and $\sqrt{\pi}$. After the gravity row emits G, the same scale is displayed as the Planck area in SI units. The pixel equation fixes this dimensionless ratio; it does not determine the SI scale by itself.

Why this step is needed: α has no units. A unitless electromagnetic number has to be matched to a unitless geometric number. Dividing the cell area by the emitted scale area gives that number without using the pixel equation to derive G.

6 Step 3: The Golden-Ratio Balance

The self-similar balance point is

$$\varphi = 1 + \frac{1}{\varphi}, \qquad \varphi^2 - \varphi - 1 = 0, \qquad \varphi = \frac{1+\sqrt{5}}{2}. \tag{3}$$

Informally: φ is the fixed point of the simplest self-similar split, where the whole-to-large ratio equals the large-to-small ratio. OPH uses it as the zero-detuning balance point for the local pixel.

The pixel does not sit exactly at φ. It sits at

$$\Delta_P := P - \varphi. \tag{4}$$

Informally: Δ_P is the small amount by which the realized cell is displaced from exact self-similar equilibrium. That small displacement is what becomes the electromagnetic observation strength.

Why this step is needed: a fixed point needs a reference position. The golden ratio supplies the self-similar reference, while $P - \varphi$ measures how far the actual cell sits away from it.

7 Step 4: Boundary Gaussian Normalization

The boundary normalization converts Δ_P into the outside coupling readout:

$$\alpha_{\text{ext}}(P) := \frac{P - \varphi}{\sqrt{\pi}}. \tag{5}$$

Informally: the screen displacement is not itself the electromagnetic coupling. The boundary Gaussian width supplies the normalization. Dividing by $\sqrt{\pi}$ turns the geometric displacement into a dimensionless observation strength.

Why this step is needed: the outside screen variable is an area displacement. The inside electromagnetic variable is a coupling. The boundary normalization is the conversion factor between those two readings.

8 Step 5: The Inside Electromagnetic Readout

Let

$$A_{\text{Th}}(P) := \alpha_{\text{em}}^{-1}(0; P) \tag{6}$$

be the inverse electromagnetic coupling at zero momentum, emitted from the same trial pixel P. The inside coupling is

$$\alpha_{\text{in}}(P) := \frac{1}{A_{\text{Th}}(P)}. \tag{7}$$

Informally: an observer inside the encoded world does not see P as a screen area. The observer sees the strength of electromagnetism. Since particle physicists quote the low-energy coupling as an inverse number near 137, the actual coupling is $1/A_{\text{Th}}(P)$.

Why this step is needed: the measured fine-structure constant is the Thomson-limit coupling. The calculation must therefore end at zero momentum, after the electroweak anchor has been transported through all charged degrees of freedom.

9 Step 6: The Fixed-Point Equation

Closure requires the outside and inside couplings to agree:

$$\alpha_{\text{ext}}(P) = \alpha_{\text{in}}(P). \tag{8}$$

Using Eqs. (5) and (7),

$$\frac{P - \varphi}{\sqrt{\pi}} = \frac{1}{A_{\text{Th}}(P)}. \tag{9}$$

Equivalently,

$$\boxed{H(P) := P - \varphi - \frac{\sqrt{\pi}}{A_{\text{Th}}(P)} = 0} \tag{10}$$

or

$$\boxed{P = G(P) := \varphi + \frac{\sqrt{\pi}}{A_{\text{Th}}(P)}.} \tag{11}$$

Informally: feed a trial pixel into the source chain. It emits a Thomson endpoint. That endpoint tells the pixel what size it should have. The physical pixel is the value that comes back unchanged.

Why this step is needed: a trial value is not enough. The same cell must agree with itself when read from the outside and from the inside. That is why the answer is a fixed point, not a one-way evaluation.

10 Step 7: The Source Map

For a trial P, the source map first emits

$$M_U(P) = E_P \, \mathrm{e}^{-2\pi} \, P^{1/6}, \tag{12}$$

$$E_{\text{cell}}(P) = \frac{E_P}{\sqrt{P}}. \tag{13}$$

Informally: $M_U(P)$ is the high-scale source readout, and $E_{\text{cell}}(P)$ is the local cell energy. Both come from the same screen cell. The unification scale is therefore read from the pixel.

Why this step is needed: the inside electromagnetic coupling has to be generated from the same P used in the outside equation. These two formulas begin that inside calculation without adding a separate high-scale fit parameter.

11 Step 8: Electroweak Transmutation

The source branch uses

$$\beta_{\text{EW}} = N_c + 1 = 4 \tag{14}$$

with $N_c = 3$, and defines

$$v(P, \alpha_U) = E_{\text{cell}}(P) \exp\left[-\frac{2\pi}{\beta_{\text{EW}} \alpha_U}\right]. \tag{15}$$

Informally: the weak scale is exponentially lower than the cell scale. The unified coupling controls that descent, so changing α_U changes the electroweak scale sharply.

Why this step is needed: the fine-structure constant is measured at low energy, but the source map begins at the cell scale. The transmutation formula explains how the electroweak scale is produced from the same source data.

12 Step 9: One-Loop Gauge Running

For $i = 1, 2, 3$, define

$$(b_1, b_2, b_3) = \left(\frac{33}{5},\ 1,\ -3\right), \tag{16}$$

and run the couplings by

$$\alpha_i^{-1}(\mu; P, \alpha_U) = \alpha_U^{-1} + \frac{b_i}{2\pi} \log\left(\frac{M_U(P)}{\mu}\right). \tag{17}$$

Informally: once α_U is guessed, all three gauge couplings at lower scales are fixed. The same high-scale coupling determines α_1, α_2, and α_3.

Why this step is needed: the electromagnetic coupling is a mixture of the weak and hypercharge couplings. Running the gauge couplings supplies the ingredients needed to form that mixture at the Z-scale.

13 Step 10: The Self-Consistent Z-Scale

The hypercharge coupling in Standard Model normalization is

$$\alpha_Y(\mu; P) = \frac{3}{5}\alpha_1(\mu; P). \tag{18}$$

The tree-level Z-mass readout is

$$m_Z(\mu; P, \alpha_U) = \frac{v(P, \alpha_U)}{2}\sqrt{4\pi\alpha_2(\mu; P, \alpha_U) + 4\pi\alpha_Y(\mu; P, \alpha_U)}. \tag{19}$$

The source point uses the self-consistency condition

$$\mu = m_Z(\mu; P, \alpha_U). \tag{20}$$

Informally: the Z-scale is determined inside the calculation. It is the scale at which the running couplings and the electroweak transmutation formula reproduce their own Z-mass readout.

Why this step is needed: using a measured m_Z here would hide experimental input inside the source map. The self-consistency condition keeps the source calculation closed.

14 Step 11: Heat-Kernel Gauge Closure

For SU(2), the irreducible representations are labeled by $j = n/2$, $n = 0, 1, \ldots, N_2$, with

$$d_j = 2j + 1, \qquad C_2(j) = j(j+1). \tag{21}$$

For SU(3), representations are labeled by highest weights (p, q), $0 \le p, q \le N_3$, with

$$d_{p,q} = \frac{(p+1)(q+1)(p+q+2)}{2}, \tag{22}$$

$$C_2(p, q) = \frac{p^2 + q^2 + pq + 3p + 3q}{3}. \tag{23}$$

For a compact group G in this finite representation cutoff, define

$$Z_G(t) = \sum_R d_R \, \mathrm{e}^{-tC_2(R)}, \tag{24}$$

$$\bar{\ell}_G(t) = \frac{1}{Z_G(t)} \sum_R d_R \, \mathrm{e}^{-tC_2(R)} \log d_R. \tag{25}$$

The heat-kernel source parameters are

$$t_2 = 4\pi^2 \alpha_2(m_Z; P, \alpha_U), \qquad t_3 = 4\pi^2 \alpha_3(m_Z; P, \alpha_U). \tag{26}$$

The pixel-closure equation is

$$\boxed{\bar{\ell}_{\mathrm{SU}(2)}(t_2) + \bar{\ell}_{\mathrm{SU}(3)}(t_3) = \frac{P}{4}.} \tag{27}$$

For a trial P, Eq. (27) is solved for $\alpha_U(P)$.

Informally: the representation entropy carried by the weak and color sectors must match the pixel capacity assigned to the cell. This step is what locks the unified coupling to the same P that appears in the outer equation.

Why this step is needed: α_U should not be chosen by hand. The heat-kernel closure turns the finite representation content of the gauge sectors into an equation for $\alpha_U(P)$.

15 Step 12: The Electroweak Anchor

Once $\alpha_U(P)$ and $m_Z(P)$ are solved, define

$$\alpha_Y(m_Z; P) = \frac{3}{5} \alpha_1(m_Z; P), \tag{28}$$

$$\alpha_{\mathrm{em}}(m_Z^2; P) = \left(\frac{1}{\alpha_2(m_Z; P)} + \frac{1}{\alpha_Y(m_Z; P)} \right)^{-1}. \tag{29}$$

The source-locked electroweak anchor is

$$\boxed{A_Z(P) := \alpha_{\mathrm{em}}^{-1}(m_Z^2; P).} \tag{30}$$

The weak mixing readout is

$$\sin^2\theta_W(m_Z;P)=\frac{\alpha_{\rm em}(m_Z^2;P)}{\alpha_2(m_Z;P)}. \tag{31}$$

Informally: the source map has reached the electromagnetic coupling at the electroweak anchor scale. The low-energy 1/137 number appears only after this anchor is transported to zero momentum.

Why this step is needed: $A_Z(P)$ is the clean place where the source map meets standard electroweak physics. From here the problem becomes a transport problem for the electromagnetic current.

16 Step 13: Charged Spectrum Used by the Transport

The exact one-loop transport uses $N_c=3$, $N_g=3$, and

$$\epsilon=\frac{1}{6},\qquad \delta=\frac{\beta_{\rm EW}}{2N_cN_g}=\frac{2}{9}. \tag{32}$$

Define three Koide roots

$$r_k=1+\sqrt{2}\cos\!\left(\delta+\frac{2\pi k}{3}\right),\qquad k=0,1,2, \tag{33}$$

then sort them in increasing order and write the sorted list as (r_1,r_2,r_3).

The quark exponent vectors are

$$\boldsymbol{n}_u=(2N_c,N_c,0)=(6,3,0),\qquad \boldsymbol{n}_d=(2N_c,N_c+1,N_c-1)=(6,4,2). \tag{34}$$

With $v=v(P)$,

$$m_u=\frac{v}{\sqrt{2}}\epsilon^6,\qquad m_c=\frac{v}{\sqrt{2}}\epsilon^3,\qquad m_t=\frac{v}{\sqrt{2}}, \tag{35}$$

$$m_d=\frac{v}{\sqrt{2}}\epsilon^6,\qquad m_s=\frac{v}{\sqrt{2}}\epsilon^4,\qquad m_b=\frac{v}{\sqrt{2}}\epsilon^2. \tag{36}$$

For charged leptons the exponent vector is

$$\boldsymbol{n}_e=(7,4,3). \tag{37}$$

Define

$$\log g_c=\frac{1}{3}\sum_{a=1}^{3}\log\!\left(\frac{r_a^2\sqrt{2}\,6^{n_{e,a}}}{v}\right),\qquad s_0=\mathrm{e}^{-\log g_c},\qquad s_e=s_0\,2^{1/6}. \tag{38}$$

Then

$$m_e=s_er_1^2,\qquad m_\mu=s_er_2^2,\qquad m_\tau=s_er_3^2. \tag{39}$$

Informally: this section specifies the charged masses used by the source transport. The lepton part is a clean one-loop calculation. The quark part is a perturbative continuation. The confined hadronic QCD spectral measure is a separate low-energy object.

Why this step is needed: vacuum polarization depends on charged particles. The transport step cannot be evaluated until the charged spectrum and its charges are specified.

17 Step 14: Exact One-Loop Fermion Transport

For a fermion of mass m_f, electric charge Q_f, and multiplicity N_f, define

$$K_f(Q^2; m_f, Q_f, N_f) = \frac{2N_f Q_f^2}{\pi} \int_0^1 x(1-x) \log\left(1 + \frac{Q^2 x(1-x)}{m_f^2}\right) \mathrm{d}x. \tag{40}$$

The integral has the following closed form. Let

$$z = \frac{Q^2}{m_f^2}, \qquad a = \frac{z}{4}. \tag{41}$$

Then

$$\int_0^1 x(1-x) \log(1 + zx(1-x))\, \mathrm{d}x = -\frac{5}{18} + \frac{1}{6a} + \frac{(2a-1)\sqrt{1+a}\ \mathrm{asinh}(\sqrt{a})}{6a^{3/2}}, \tag{42}$$

with

$$\mathrm{asinh}(\sqrt{a}) = \log(\sqrt{a} + \sqrt{1+a}). \tag{43}$$

The lepton contribution is

$$\Delta_{\text{lep}}(P) = K_e(m_Z(P)^2; m_e, 1, 1) + K_\mu(m_Z(P)^2; m_\mu, 1, 1) + K_\tau(m_Z(P)^2; m_\tau, 1, 1). \tag{44}$$

The naive five-quark contribution is

$$\begin{aligned}\Delta_q^{\text{naive}}(P) = {} & K_u(m_Z(P)^2; m_u, 2/3, 3) + K_d(m_Z(P)^2; m_d, -1/3, 3) \\ & + K_s(m_Z(P)^2; m_s, -1/3, 3) + K_c(m_Z(P)^2; m_c, 2/3, 3) + K_b(m_Z(P)^2; m_b, -1/3, 3).\end{aligned} \tag{45}$$

The perturbative screening factor is

$$S_{\text{calc}}(P) = 1 - \frac{N_c \alpha_3(m_Z; P)}{\pi}. \tag{46}$$

Thus

$$\Delta_{\text{calc}}(P) = \Delta_{\text{lep}}(P) + S_{\text{calc}}(P)\Delta_q^{\text{naive}}(P). \tag{47}$$

Informally: charged leptons are treated by an exact one-loop kernel. Quarks are treated by a perturbative expression with a simple screening factor. The expected gap is the confined hadronic spectral transport plus the matching terms needed to use the same electromagnetic-current convention all the way to the endpoint.

Why this step is needed: the Thomson endpoint is not the same as the Z-scale anchor. Charged particles screen the electromagnetic current between those scales, and the kernel is the mathematical form of that screening.

18 Step 15: The Source Calculation Endpoint

The calculated endpoint is

$$A_{\text{calc}}(P) = A_Z(P) + \Delta_{\text{calc}}(P). \tag{48}$$

Putting A_{calc} into Eq. (11) gives the source map

$$G_{\text{calc}}(P) = \varphi + \frac{\sqrt{\pi}}{A_{\text{calc}}(P)}. \tag{49}$$

The numerical solve uses

$$G_{\text{calc}}(P_{\text{root}}) = P_{\text{root}} \tag{50}$$

and emits

$$P_{\text{root}} = 1.63097209569432901817967892561191884270169, \tag{51}$$

$$\alpha_{\text{root}}^{-1} = 136.994835164621649457949994585787193262029. \tag{52}$$

The source anchor at that root point is

$$A_Z(P_{\text{root}}) = 128.308268045165213892552005990181778935450. \tag{53}$$

The calculated transport contribution is

$$\Delta_{\text{calc}}(P_{\text{root}}) = 8.68656711945643556539798859560541432657857. \tag{54}$$

Informally: the fixed-point algebra lands at a stable value near 137, with no measured alpha inserted into the solve. The remaining gap has a precise address: low-energy hadronic transport in the same endpoint convention.

Why this step is needed: this is the non-circular check. It shows what the OPH source chain produces before the calibrated hadronic endpoint correction is added.

19 Step 16: The CODATA/NIST Measured Endpoint and Comparison Pixel

The CODATA/NIST 2022 inverse fine-structure constant is

$$A_{\text{C}} = 137.035999177, \qquad \sigma_A = 0.000000021, \tag{55}$$

with concise form 137.035999177(21). The corresponding comparison pixel is

$$P_{\text{C}} = \varphi + \frac{\sqrt{\pi}}{A_{\text{C}}} = \begin{array}{l}\texttt{1.6309682094039593248792798477826489413359828516279250606661}\\ \texttt{507533907793398933432}\end{array} \tag{56}$$

The CODATA/NIST central coupling is

$$\alpha(0) = \frac{1}{A_{\text{C}}} = \begin{array}{l}\texttt{0.0072973525643314250302457952646916832280660213133653604957}\\ \texttt{798803819933561573639928}\end{array} \tag{57}$$

Informally: once the measured Thomson endpoint is supplied, the outer OPH equation fixes the corresponding comparison pixel immediately. It is the direct readout of one equation.

Why this step is needed: the measured endpoint and the comparison pixel are two forms of the same fixed-point statement. Reporting both makes it clear how the measured inverse coupling maps back to the screen cell.

Two-P provenance audit. There are two distinct pixel numerals in this paper. The zero-input source solve gives P_{root} in Eq. (51); it is the branch used to audit what the OPH source chain emits before a same-scheme hadronic endpoint correction is supplied. The comparison value P_{C} in Eq. (56) is different by about 3.9×10^{-6} in P-space because it is defined from the measured CODATA/NIST Thomson endpoint. Thus the published comparison pixel is a measured-endpoint display, not a derivation of α. Downstream quantities that use P_{C} inherit that measured endpoint by definition; downstream quantities that use P_{root} are source-branch diagnostics.

20 Step 17: Endpoint Accounting at the CODATA/NIST Comparison Pixel

At $P = P_{\text{C}}$, the source point gives

$$A_Z(P_{\text{C}}) = \begin{array}{l}\texttt{128.30796547328624820996110874175671618724547618036535646005}\\\texttt{342169635117784168285644078724728}\end{array} \tag{58}$$

$$\Delta_{\text{calc}}(P_{\text{C}}) = \begin{array}{l}\texttt{8.686567842708528400985442542885969768267221737648736423378 4}\\\texttt{5773899937844591067830809611795 90}\end{array} \tag{59}$$

The transport required by the CODATA/NIST measured endpoint is

$$\Delta_{\text{req}}(P_{\text{C}}) = A_{\text{C}} - A_Z(P_{\text{C}}) = \begin{array}{l}\texttt{8.728033703713751790038891258243283812754523819634643539946 5}\\\texttt{783036488221583171435592127527 2}\end{array} \tag{60}$$

Therefore the source-side residual at the CODATA/NIST comparison pixel is

$$\begin{aligned} R_Q(P_{\text{C}}) &= \Delta_{\text{req}}(P_{\text{C}}) - \Delta_{\text{calc}}(P_{\text{C}}) \\ &= \begin{array}{l}\texttt{0.041465861005223389053448715357314044487302081985907116568 1}\\\texttt{2056464944371240646525111663476}\end{array} \end{aligned} \tag{61}$$

Informally: the CODATA/NIST measured value differs from the pure source calculation by a small, precisely localized inverse-alpha contribution. The golden-ratio equation, the heat-kernel closure, and the numerical fixed-point solve all point to the same address for the difference: the same-scheme low-energy hadronic transport.

Why this step is needed: this accounting prevents the hadronic contribution from being hidden inside the final number. It shows the source anchor, the calculated transport, the required transport, and the residual in the same units.

21 Step 18: The OPH Output and the Required Hadronic Correction

The fixed-point calculation first emits

$$A_{\text{calc}}(P_{\text{root}}) = \alpha_{\text{root}}^{-1} = \texttt{136.99483516462164945794999458578719326 2029} \quad . \tag{62}$$

The finite-screen unified gauge-width contribution is

$$\alpha_U(P_\star) = 0.041124336195630495.$$

Adding that emitted gauge-width contribution gives the current OPH no-hadron output

$$A^{\mathrm{fp}}_{\alpha_U} = \alpha^{-1}_{\mathrm{root}} + \alpha_U(P_\star) = 137.035959500817279952949994585787\ldots, \tag{63}$$

or

$$\alpha^{\mathrm{fp}}_{\mathrm{src}+U} = \left(A^{\mathrm{fp}}_{\alpha_U}\right)^{-1} = 0.00729735467714250592971405837329\ldots. \tag{64}$$

This $A^{\mathrm{fp}}_{\alpha_U}$ value is the numerical output of the OPH fine-structure calculation in this paper. It is the source-side prediction before the missing Ward-projected hadronic spectral transport is supplied.

Compared with the CODATA/NIST central measured endpoint,

$$A_{\mathrm{C}} = 137.035999177, \qquad \sigma_A = 0.000000021, \tag{65}$$

the required same-scheme hadronic endpoint correction is

$$\Delta^{\mathrm{fp}}_{\mathrm{H,req}} = A_{\mathrm{C}} - A^{\mathrm{fp}}_{\alpha_U} = \mathtt{0.000039676182720047050005414213}\ldots. \tag{66}$$

Equivalently,

$$\Delta^{\mathrm{fp}}_{\mathrm{H,req}} = 0.0000396762 \pm 0.000000021, \qquad \frac{\Delta^{\mathrm{fp}}_{\mathrm{H,req}}}{A_{\mathrm{C}}} = 2.89531 \times 10^{-7}. \tag{67}$$

This is the inverse-alpha gap between the current OPH output and the CODATA/NIST central value. Its physical address is the low-energy QCD/hadronic vacuum-polarization and same-scheme transport of the Ward-projected electromagnetic current.

For bookkeeping, the full root-to-CODATA difference is

$$\begin{aligned}\Delta^{\mathrm{fp}}_{\mathrm{H,cal}} &= A_{\mathrm{C}} - A_{\mathrm{calc}}(P_{\mathrm{root}}) \\ &= \mathtt{0.041164012378350542050005414212806737971} \\ &= \alpha_U(P_\star) + \Delta^{\mathrm{fp}}_{\mathrm{H,req}}.\end{aligned} \tag{68}$$

This larger difference is bookkeeping: it contains both the emitted finite-screen unified gauge-width contribution and the remaining hadronic correction.

Equivalently, the CODATA/NIST central value can be written as

$$A_{\mathrm{C}} = \alpha^{-1}_{\mathrm{root}} + \alpha_U(P_\star)\, C_{24,Q}, \qquad C_{24,Q} = 1.00096478597323262538495111407024 75\ldots. \tag{69}$$

The factor $C_{24,Q}$ is a compact accounting of the finite same-scheme hadronic endpoint transport needed for source-only closure. Equivalently, $\Delta^{\mathrm{fp}}_{\mathrm{H,req}} = \alpha_U(P_\star)(C_{24,Q} - 1)$. The 24-register argument emits $\alpha_U(P_\star)$; the correction factor is the remaining hadronic/QCD payload.

Here A_{C} is the CODATA/NIST 2022 inverse fine-structure constant [19, 20]. The source value $A_{\mathrm{calc}}(P_{\mathrm{root}}) = \alpha^{-1}_{\mathrm{root}}$ and the source-plus-unified-gauge-width value $A^{\mathrm{fp}}_{\alpha_U}$ are reproducible OPH fixed-point outputs [4]. Equation (66) is the measured-comparison residual in inverse-alpha units.

Informally: the OPH calculation outputs 137.035959500817…. The CODATA/NIST measured value is 137.035999177(21). The difference is a small number with a known physical meaning: the hadronic/QCD contribution in the same electromagnetic-current convention. The open task is to compute that hadronic piece from source data.

This marks the calculation boundary: additional digits require a source-derived hadronic spectral calculation.

The endpoint decomposition is

$$A_{\mathrm{Th}}(P) = A_Z(P) + \Delta_{\mathrm{lep}}(P) + \Delta_{\mathrm{had}}(P) + \Delta_{\mathrm{EW}}(P). \tag{70}$$

The calculated transport contains the exact one-loop charged-lepton contribution and a structured quark-screening continuation. The remaining pure source object can be written compactly as

$$R_Q(P) = \Delta_{\mathrm{had}}(P) + \Delta_{\mathrm{EW}}(P) - \left[\Delta_{\mathrm{calc}}(P) - \Delta_{\mathrm{lep}}(P)\right], \tag{71}$$

where the right side is understood in the same renormalization and endpoint scheme as $A_Z(P)$.

In the screening notation used by the endpoint calculation, let

$$x(P) = \frac{N_c \alpha_3(m_Z; P)}{\pi}. \tag{72}$$

At $P = P_{\mathrm{C}}$,

$$S_{\mathrm{required}} = \begin{array}{l}\texttt{0.89540013264765879780580028318167064130770986481229164844 83}\\ \texttt{059101154524252127308688810786457 6}\end{array} \tag{73}$$

$$c_Q = \begin{array}{l}\texttt{0.6580257599271554356382301702323600504249200990705863960856}\\ \texttt{6007832470711257322011342309305088}\end{array} \tag{74}$$

where

$$S_{\mathrm{required}} = 1 - x + c_Q x^2. \tag{75}$$

Informally: the needed hadronic and endpoint correction can be summarized as a small second-order screening coefficient. A pure source proof has to emit this coefficient from a Ward-projected hadronic spectral measure.

Why this step is needed: hadrons are not pointlike free quarks at low energy. The coefficient is a compact way to record what the empirical hadronic spectral function contributes in this endpoint convention.

22 Step 19: Why Raw PDG $\Delta\alpha^{(5)}_{\mathrm{had}}(M_Z)$ Is a Different Quantity

The direct PDG/CERN diagnostic uses

$$A_Z(P_{\mathrm{C}}) = \begin{array}{l}\texttt{128.3079654732862482099611087417567161872454761803653564600 5}\\ \texttt{342169635117784168285644}\end{array} \tag{76}$$

$$\Delta_{\mathrm{lep}}(P_{\mathrm{C}}) = \begin{array}{l}\texttt{4.3093978664522040271317438975344894018487156605576773194711}\\ \texttt{5280896656803132579064661 29}\end{array} \tag{77}$$

$$\Delta\alpha^{(5)}_{\mathrm{had}}(M_Z) = 0.02761. \tag{78}$$

It forms

$$A_L = A_Z + \Delta_{\mathrm{lep}}, \qquad A_{\mathrm{PDGdiag}} = \frac{A_L}{1 - \Delta\alpha^{(5)}_{\mathrm{had}}(M_Z)}. \tag{79}$$

Numerically,

$$A_{\mathrm{PDGdiag}} = \begin{array}{l}\texttt{136.3828950726955771214151242189771651180022335080811544539 9}\\ \texttt{95007202025379456891237945812 89}\end{array} \tag{80}$$

This differs from 137.035999177. Holding this raw hadronic denominator shift fixed, the target would require

$$\Delta\alpha_{\mathrm{had,req}} = \begin{array}{l}\texttt{0.03224434355788728882226849923695794742200734761222608150 65}\\\texttt{930892218713809261179501414 38549}\end{array} \tag{81}$$

or a same-scheme source-anchor bridge of

$$\begin{array}{l}\texttt{0.6350718999845777629071473607087944109058081590769662204754}\\\texttt{254946822541269913529133871}\end{array} \tag{82}$$

inverse-alpha units.

Informally: the raw electroweak-review hadronic running number is a useful diagnostic. The OPH endpoint needs the same electromagnetic current and the same inverse-alpha endpoint convention used by $A_Z(P)$. Mixing the two conventions moves the answer by a visible amount.

Why this step is needed: a common mistake is to insert a published $\Delta\alpha^{(5)}_{\mathrm{had}}(M_Z)$ number as though it were the OPH endpoint contribution. This section shows the numerical consequence of that convention mismatch.

23 Step 20: The Measured Hadronic Spectral Input

The standard measured hadronic vacuum-polarization relation uses

$$\Delta\alpha_{\mathrm{had}}(q^2) = -\frac{\alpha q^2}{3\pi}\,\mathrm{P.V.}\int \frac{R(s)}{s(s-q^2)}\,\mathrm{d}s, \tag{83}$$

where $R(s)$ is the measured ratio of the bare $e^+e^- \to$ hadrons cross section to the pointlike $e^+e^- \to \mu^+\mu^-$ cross section.

The independently documented measured quantity in the standard electroweak literature is usually quoted as $\Delta\alpha^{(5)}_{\mathrm{had}}(M_Z^2)$. It is a dimensionless change in the running electromagnetic coupling at the Z-boson mass, not the inverse-alpha correction $\Delta^{\mathrm{fp}}_{\mathrm{H,req}}$ used in Eq. (66). For example, Davier, Hoecker, Malaescu, and Zhang report

$$\Delta\alpha^{(5)}_{\mathrm{had}}(M_Z^2) = (275.7 \pm 1.0) \times 10^{-4}$$

from e^+e^--based data [12]. A more recent perturbative update by Erler and Ferro-Hernandez gives

$$\Delta\alpha^{(5)}_{\mathrm{had}}(M_Z^2) = (276.29 \pm 0.38 \pm 0.62) \times 10^{-4}$$

when low-energy cross-section data are used as input [15].

The comparison residual in Eq. (66) is different. It is expressed in inverse-alpha units and in the endpoint convention used by $A_Z(P)$. The standard published hadronic-running values document the physics source of the correction. They do not by themselves give the OPH same-scheme inverse-alpha correction to arbitrary precision. The community does publish the data machinery needed for this kind of calculation. Jegerlehner's alphaQED package provides hadronic running routines, covariance data, and an integration routine for custom kernels [17]. That is the right empirical route once the OPH endpoint kernel and finite same-scheme remainder are fixed.

In the OPH endpoint convention this is represented as a same-current spectral functional,

$$\Delta_{\mathrm{had}}(P) = \frac{m_Z(P)^2}{3\pi} \int \frac{\rho_Q(s;P)}{s[s+m_Z(P)^2]}\,\mathrm{d}s, \tag{84}$$

plus the same-scheme finite remainder needed by Eq. (70).

Informally: the measured hadronic input supplies the electromagnetic spectral information required for the endpoint. The single published number 0.0276 is one weighted integral of that spectral information. The OPH endpoint needs a different weighted integral plus the same-scheme finite remainder. A pure source theorem requires the same spectral object from OPH source data.

Why this step is needed: the dispersion relation explains why measured $e^+e^- \to$ hadrons data are the right empirical input. They measure the electromagnetic spectral function that the endpoint transport requires.

24 No-Go: The 24-Register Scale Does Not Determine the Hadronic Residual

Theorem 1 (Coarse 24-register data do not determine the Thomson residual)**.** *The specified OPH source objects P, $\alpha_U(P)$, $N_c = 3$, $N_g = 3$, and the 24-slot repair-register count do not determine the exact low-energy Thomson residual.*

Proof. The hadronic endpoint contribution depends on the Ward-projected spectral functional

$$\Delta_{\mathrm{had}}(P) = \frac{m_Z(P)^2}{3\pi} \int_0^\infty \frac{\rho_Q(s;P)}{s[s+m_Z(P)^2]}\,\mathrm{d}s \tag{85}$$

up to the same-scheme finite endpoint remainder. Let

$$K_P(s) = \frac{m_Z(P)^2}{3\pi s[s+m_Z(P)^2]}.$$

This kernel is not constant. Two positive spectral measures can therefore have the same coarse register count, the same leading normalization, and the same electroweak branch data while giving different residuals. For example, with $s_1 \neq s_2$,

$$\rho_1(s) = M\delta(s-s_1), \qquad \rho_2(s) = M\delta(s-s_2)$$

have equal total mass, but

$$\int K_P(s)\rho_1(s)\,\mathrm{d}s = MK_P(s_1), \qquad \int K_P(s)\rho_2(s)\,\mathrm{d}s = MK_P(s_2),$$

which are generically unequal. Thus the 24-register scale explains the size of the correction but does not select the spectral measure that fixes its exact value. □

Theorem 2 (Tautological 24-Jacobi realization)**.** *For any positive residual $R > 0$, there is an exact 24-point Jacobi/Stieltjes representation that reproduces R. Such a representation is non-predictive unless its nodes, weights, normalization, and finite remainder are emitted by OPH source rules before comparison with the Thomson endpoint.*

Proof. Choose any positive spectral nodes $s_1, \ldots, s_{24} > 0$ and any positive normalized weights $u_j > 0$ with $\sum_j u_j = 1$. The finite inverse Stieltjes/Favard construction gives a positive 24×24 Jacobi matrix $J_{24,Q}$ whose e_1-spectral measure is

$$\mathrm{d}\mu_0(s) = \sum_{j=1}^{24} u_j \delta(s - s_j).$$

Equivalently,

$$e_1^\top f(J_{24,Q}) e_1 = \sum_{j=1}^{24} u_j f(s_j)$$

for every function f on the spectrum. Define

$$D(J_{24,Q}; m) = e_1^\top \left[J_{24,Q}^{-1} - (J_{24,Q} + m^2 I)^{-1} \right] e_1.$$

Since $s_j > 0$,

$$D(J_{24,Q}; m) = \sum_{j=1}^{24} u_j \left(\frac{1}{s_j} - \frac{1}{s_j + m^2} \right) > 0.$$

For any same-scheme finite remainder $\Xi_Q < R$, set

$$\omega_Q = \frac{3\pi(R - \Xi_Q)}{D(J_{24,Q}; m)}.$$

Then

$$\frac{\omega_Q}{3\pi} e_1^\top \left[J_{24,Q}^{-1} - (J_{24,Q} + m^2 I)^{-1} \right] e_1 + \Xi_Q = R.$$

The construction reproduces the chosen residual exactly, but the freedom in s_j, u_j, ω_Q, and Ξ_Q means it is a back-solving theorem rather than a source derivation. □

The source-side object needed for a final OPH endpoint theorem is therefore

$$R_Q(P) = \frac{\omega_Q(P)}{3\pi} e_1^\top \left[J_{24,Q}(P)^{-1} - \left(J_{24,Q}(P) + m_Z(P)^2 I\right)^{-1} \right] e_1 + \Xi_Q(P), \tag{86}$$

or equivalently the source-emitted spectral density $\rho_Q(s; P)$ plus the same-scheme finite remainder. The required dependency graph must have no path from the measured Thomson endpoint into $J_{24,Q}$, ω_Q, Ξ_Q, or ρ_Q.

25 Step 21: The Conditional Pure Source Theorem

Theorem 3 (OPH fine-structure endpoint, conditional pure source form)**.** *Assume:*

(i) the OPH overlap-consistency branch emits the source map $P \mapsto A_Z(P)$ by Eqs. (12)–(30);

(ii) the Ward-projected $\mathrm{U}(1)_Q$ transport theorem emits a same-scheme endpoint map

$$A_{\mathrm{Th}}(P) = A_Z(P) + \Delta_{\mathrm{lep}}(P) + \Delta_{\mathrm{had}}(P) + \Delta_{\mathrm{EW}}(P);$$

(iii) $G(P) = \varphi + \sqrt{\pi}/A_{\mathrm{Th}}(P)$ is a self-map and a contraction on the physical pixel interval I;

(iv) the interval image contains the root and the residual bound is certified.

Then there is a unique $P_\star \in I$ satisfying $G(P_\star) = P_\star$, and the fine-structure constant on that branch is

$$\alpha(0) = \frac{1}{A_{\mathrm{Th}}(P_\star)} = \frac{P_\star - \varphi}{\sqrt{\pi}}.$$

Proof. By assumption (iii), $G : I \to I$ is a contraction. Banach's fixed-point theorem gives a unique $P_\star \in I$ such that $G(P_\star) = P_\star$. Equation (11) gives

$$P_\star - \varphi = \frac{\sqrt{\pi}}{A_{\mathrm{Th}}(P_\star)}.$$

Dividing by $\sqrt{\pi}$ gives

$$\frac{P_\star - \varphi}{\sqrt{\pi}} = \frac{1}{A_{\mathrm{Th}}(P_\star)}.$$

The left side is $\alpha_{\mathrm{ext}}(P_\star)$, and the right side is $\alpha_{\mathrm{in}}(P_\star)$. Their common value is the Thomson-limit electromagnetic coupling $\alpha(0)$. □

Informally: a source-derived hadronic spectral endpoint map would close the last open transport step. With the interval proof included, the fine-structure constant is the unique fixed point of the full source map.

The release gate for promoting this row from empirical endpoint comparison to source-only theorem is explicit. The packet must contain

`WARD_Q_SPECTRAL_MEASURE_RECEIPT`, `J24Q_SOURCE_JACOBI_RECEIPT`, `OMEGAQ_NORMALIZATION_RECEIPT`, `XI_SAME_SCHEME_REMAINDER_RECEIPT`, `NO_THOMSON_TARGET_LEAK_DAG`, `FULL_PIXEL_CONTRACTION_INTERVAL` and `ALPHA_PREDICTION_BEATS_DIRECT_MEASUREMENT` under the direct-measurement interval adopted for that release. For the present comparison standard, the direct rubidium recoil result of Morel, Yao, Cladé, and Guellati-Khélifa reports $\alpha^{-1} = 137.035999206(11)$ with 81 parts-per-trillion relative precision [14]. The operational receipt is therefore

$$\mathrm{rad}(\alpha^{-1}_{\mathrm{pred}}) < 1.1 \times 10^{-8}$$

in inverse-alpha units, unless a later release updates the adopted direct-measurement interval.

26 Step 22: The CODATA/NIST Comparison Endpoint

With the CODATA/NIST measured endpoint value

$$A_{\mathrm{Th}}(P_{\mathrm{C}}) = A_{\mathrm{C}} = 137.035999177(21), \tag{87}$$

Eq. (11) gives the corresponding comparison pixel:

$$\boxed{P_{\mathrm{C}} = \begin{array}{l}\texttt{1.630968209403959324879279847782648941335982851627925060666 1}\\ \texttt{507533907793398933432}\end{array}} \tag{88}$$

$$\boxed{\alpha(0) = \begin{array}{l}\texttt{0.0072973525643314250302457952646916832280660213133653604957}\\ \texttt{7988038199335615736399 28}\end{array}} \tag{89}$$

$$\boxed{\alpha^{-1}(0) = 137.035999177(21).} \tag{90}$$

Informally: the endpoint number is the same number NIST/CODATA reports. The derivation exposes every OPH-side step and identifies where the low-energy hadronic calculation enters. The missing hadronic spectral calculation is work in progress.

27 Reproducibility

The executable derivation lives in the public repository under https://github.com/FloatingPragma/observer-patch-holography/tree/main/code/P_derivation. The main human-facing CLI is

```
cd reverse-engineering-reality/code/P_derivation
python3 derive_p.py --color always
```

To print only the pure source value:

```
python3 derive_p.py --no-hadron-closure
```

To emit machine-readable output:

```
python3 derive_p.py --json --output runtime/report.json
```

To run the raw PDG/CERN diagnostic discussed in Section 19:

```
python3 fine_structure_fixed_point_demo.py --compare-alpha-inv 137.035999177
```

Informally: the command line separates the pure source value, the CODATA/NIST comparison value, and the diagnostic values. The mathematical paper uses the same separation.

28 Checklist of Non-Omitted Steps

Step	Mathematical object	Status
Overlap rule	Eq. (1)	OPH starting structure.
Pixel variable	$P = a_{\text{cell}}/\ell_\star^2$	Defined.
Golden balance	$\varphi = (1+\sqrt{5})/2$	Defined.
Boundary width	$\alpha_{\text{ext}} = (P-\varphi)/\sqrt{\pi}$	Defined.
Inside readout	$\alpha_{\text{in}} = 1/A_{\text{Th}}(P)$	Defined.
Closure	$P = \varphi + \sqrt{\pi}/A_{\text{Th}}(P)$	Defined.
Source scale	$M_U = E_P e^{-2\pi} P^{1/6}$	Defined and calculated.
Cell scale	$E_{\text{cell}} = E_P/\sqrt{P}$	Defined and calculated.
Transmutation	$v = E_{\text{cell}} \exp[-2\pi/(\beta_{\text{EW}}\alpha_U)]$	Defined and calculated.
Gauge running	Eq. (17)	Defined and calculated.

Step	Mathematical object	Status
Z-scale self-consistency	Eq. (20)	Defined and calculated.
Heat-kernel closure	Eq. (27)	Defined and calculated.
Electroweak anchor	$A_Z = \alpha_{\text{em}}^{-1}(m_Z^2; P)$	Defined and calculated.
Charged spectrum	Eqs. (36) and (39)	Calculated continuation.
Fermion kernel	Eqs. (40) and (42)	Exact one-loop kernel.
Calculated endpoint	$A_{\text{calc}} = A_Z + \Delta_{\text{calc}}$	Raw no-hadron source value.
Bare-α_U source prediction	$A^{\text{fp}}_{\alpha_U}$, Eq. (63)	Source-side no-hadron prediction.
Required hadronic correction	$\Delta^{\text{fp}}_{\text{H,req}}$, Eq. (66)	CODATA/NIST comparison residual attributed to the missing same-scheme hadronic endpoint transport.
Hadronic/scheme residual	$R_Q = A_{\text{Th}} - A_{\text{calc}}$	Isolated endpoint contribution.
CODATA/NIST endpoint	$A_{\text{Th}} = 137.035999177(21)$	Measured endpoint comparison value.
Two-P provenance	P_{root} versus P_{C}	Source-branch root and measured-endpoint comparison pixel are distinct; P_{C} inherits CODATA/NIST by definition.
Raw PDG diagnostic	Eq. (79)	Comparison diagnostic in a different endpoint convention.
Interval theorem	Banach/contraction certificate for full G	Required for pure source theorem.

29 Conclusion

The fine-structure constant is the coupling that makes one OPH screen cell internally consistent. The outside reading says that the cell is displaced from the golden-ratio balance by $(P - \varphi)/\sqrt{\pi}$. The inside reading sends the same P through the source map, gauge closure, electroweak anchor, and Ward-projected electromagnetic transport, producing $1/A_{\text{Th}}(P)$. The physical cell is the fixed point where those two readings agree.

The pure source calculation computes the root witness

$$\alpha_{\mathrm{root}}^{-1} = \texttt{136.99483516462164945794999458578719326202}9 \qquad .$$

The CODATA/NIST measured endpoint value is

$$\alpha^{-1}(0) = 137.035999177(21),$$

with

$$P = 1.63096820940395932487927984778264894 1\ldots.$$

Adding the emitted finite-screen unified gauge-width contribution gives the source-side no-hadron near-endpoint,

$$A_{\alpha_U}^{\mathrm{fp}} = 137.0359595008172799529499945857 87\ldots.$$

The remaining gap is a named low-energy hadronic and same-scheme electromagnetic contribution. The CODATA/NIST exact-value row is a measured comparison. The pure source theorem requires the Ward-projected hadronic spectral measure, the same-scheme bridge, and the interval-certificate step.

Declarations

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Code Availability declaration. The fixed-point code and runtime artifacts are available at `https://github.com/FloatingPragma/observer-patch-holography/tree/main/code/P_derivation`.

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References

[1] B. Müller, A. Osika, M. Poneder, K. Xue, B. Cassie, P. Nguyen, M. A. Visser, K. A. Anirudha, D. Matscheko, and J. Hill, *Observers Are All You Need*. GitHub source: `https://github.com/FloatingPragma/observer-patch-holography/blob/main/paper/observers_are_all_you_need.tex`. GitHub PDF: `https://github.com/FloatingPragma/observer-patch-holography/blob/main/paper/observers_are_all_you_need.pdf`.

[2] B. Müller, A. Osika, M. Poneder, K. Xue, P. Nguyen, and M. A. Visser, *Recovering Relativity and the Standard Model from Observer Overlap Consistency*. GitHub source: https://github.com/FloatingPragma/observer-patch-holography/blob/main/paper/recovering_relativity_and_standard_model_structure_from_observer_overlap_consistency_compact.tex. GitHub PDF: https://github.com/FloatingPragma/observer-patch-holography/blob/main/paper/recovering_relativity_and_standard_model_structure_from_observer_overlap_consistency_compact.pdf.

[3] B. Müller, A. Osika, M. Poneder, K. Xue, and M. A. Visser, *Deriving the Particle Zoo from Observer Consistency*. GitHub source: https://github.com/FloatingPragma/observer-patch-holography/blob/main/paper/deriving_the_particle_zoo_from_observer_consistency.tex. GitHub PDF: https://github.com/FloatingPragma/observer-patch-holography/blob/main/paper/deriving_the_particle_zoo_from_observer_consistency.pdf.

[4] OPH executable derivation code, https://github.com/FloatingPragma/observer-patch-holography/tree/main/code/P_derivation. Primary files: `paper_math.py`, `derive_p.py`, `fine_structure_fixed_point_demo.py`.

[5] OPH hadron data policy, https://github.com/FloatingPragma/observer-patch-holography/blob/main/HADRON.md.

[6] U. D. Jentschura and I. Nandori, *Attempts at a determination of the fine-structure constant from first principles: A brief historical overview*, Eur. Phys. J. H **39**, 591–613 (2014). https://arxiv.org/abs/1411.4673.

[7] R. Heyrovska and S. Narayan, *Fine-structure Constant, Anomalous Magnetic Moment, Relativity Factor and the Golden Ratio that Divides the Bohr Radius*. https://arxiv.org/abs/physics/0509207.

[8] T. P. Singh, *Quantum theory without classical time: octonions, and a theoretical derivation of the fine structure constant* 1/137, Int. J. Mod. Phys. D **30**, 2142010 (2021). https://arxiv.org/abs/2110.07548.

[9] J. Ciborowski, *Bi-Constructible pattern of weak and flavour mixing: implications for electroweak coupling constants*. https://arxiv.org/abs/2508.00030.

[10] S. V. Chekanov and H. Kjellerstrand, *Discovering the Underlying Analytic Structure Within Standard Model Constants Using Artificial Intelligence*, Particles **8**, 95 (2025). https://arxiv.org/abs/2507.00225.

[11] Q. Liu, I. Low, and Z. Yin, *A Quantum Computational Determination of the Weak Mixing Angle in the Standard Model*. https://arxiv.org/abs/2509.18251.

[12] M. Davier, A. Hoecker, B. Malaescu, and Z. Zhang, *Reevaluation of the Hadronic Contributions to the Muon $g-2$ and to $\alpha(M_Z)$*, Eur. Phys. J. C **71**, 1515 (2011). https://arxiv.org/abs/1010.4180.

[13] A. Keshavarzi, D. Nomura, and T. Teubner, *The $g-2$ of charged leptons, $\alpha(M_Z^2)$ and the hyperfine splitting of muonium*, Phys. Rev. D **101**, 014029 (2020). https://arxiv.org/abs/1911.00367.

[14] L. Morel, Z. Yao, P. Cladé, and S. Guellati-Khélifa, *Determination of the fine-structure constant with an accuracy of 81 parts per trillion*, Nature **588**, 61–65 (2020). https://doi.org/10.1038/s41586-020-2964-7.

[15] J. Erler and R. Ferro-Hernandez, *Perturbative contributions to* $\Delta\alpha^{(5)}(M_Z^2)$. https://arxiv.org/abs/2308.05740.

[16] Particle Data Group, *Data files and plots of cross-sections and related quantities in the Review of Particle Physics*. https://pdg.lbl.gov/current/xsect/.

[17] F. Jegerlehner, *Software packages alphaQED and pQCDAdler*. https://people.physik.hu-berlin.de/~fjeger/software.html.

[18] B. Müller, *Fine-Structure Constant from First Principles*, Floating Pragma Blog. https://blog.floatingpragma.io/fine-structure-constant-from-first-principles.

[19] NIST, *CODATA Value: inverse fine-structure constant*, 2022 CODATA recommended value. https://physics.nist.gov/cgi-bin/cuu/Value?eqalphinv=.

[20] P. J. Mohr, E. Tiesinga, D. B. Newell, and B. N. Taylor, *CODATA Recommended Values of the Fundamental Physical Constants: 2022*. NIST publication page: https://www.nist.gov/publications/codata-recommended-values-fundamental-physical-constants-2022.

[21] NIST/CODATA, *CODATA Recommended Values of the Fundamental Physical Constants: 2022*, wall chart. https://physics.nist.gov/cuu/pdf/wall_2022.pdf.